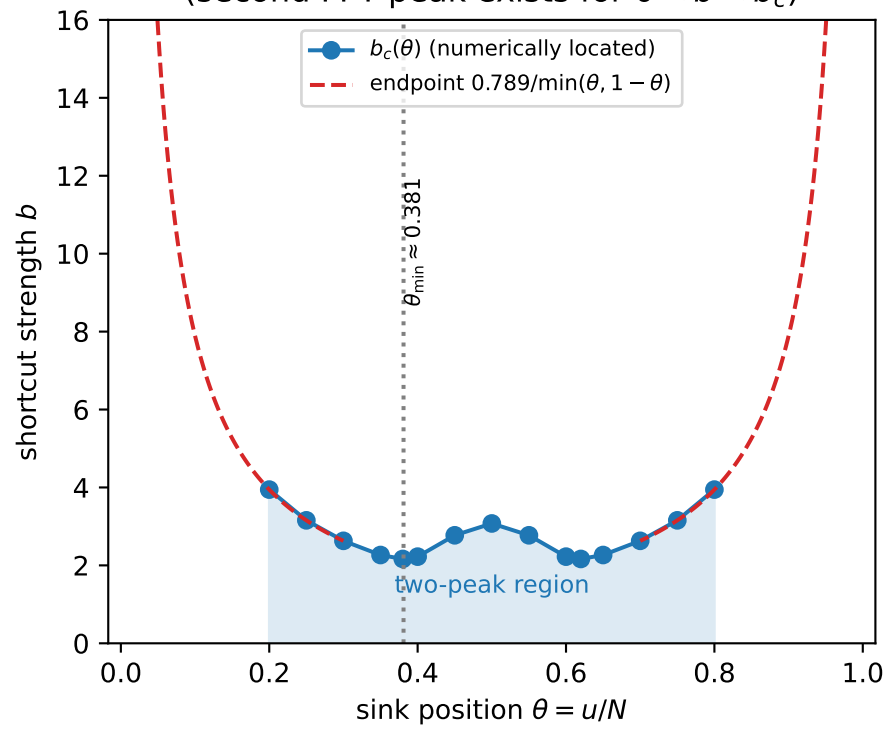
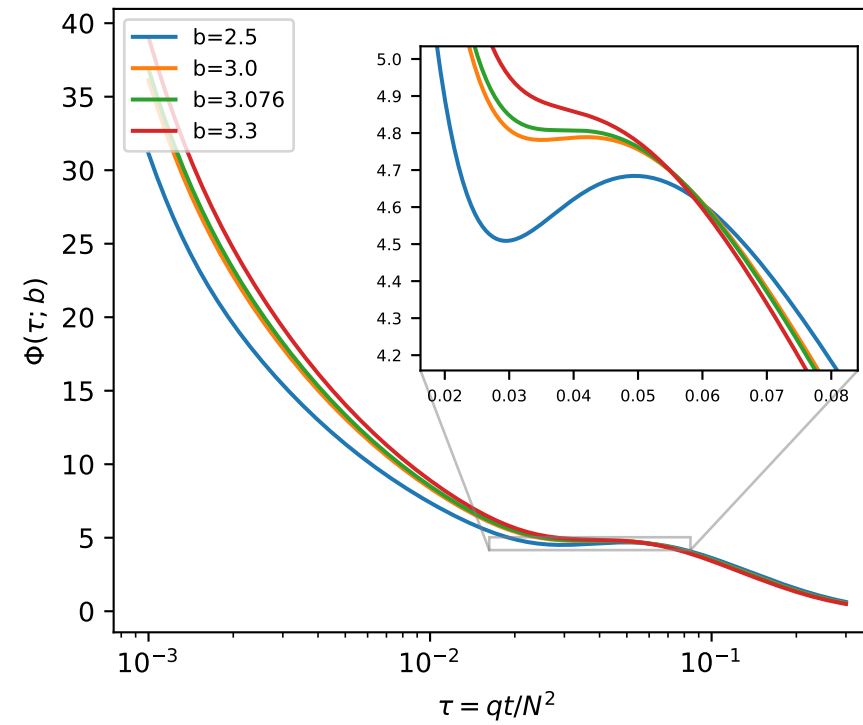


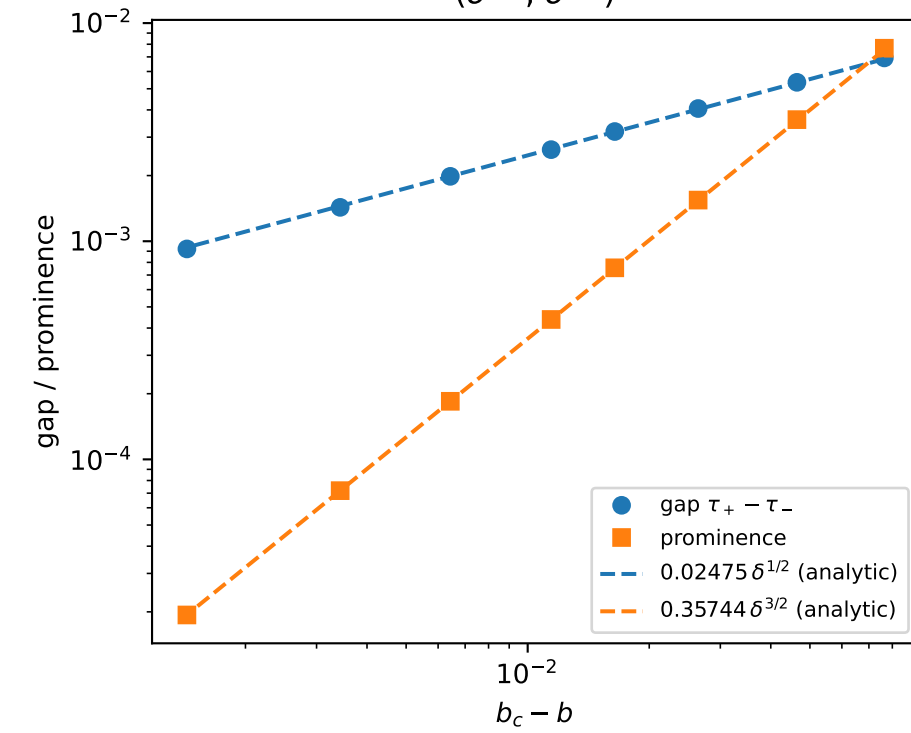
(a) saddle-node phase boundary $b_c(\theta)$
(second FPT peak exists for $0 < b < b_c$)



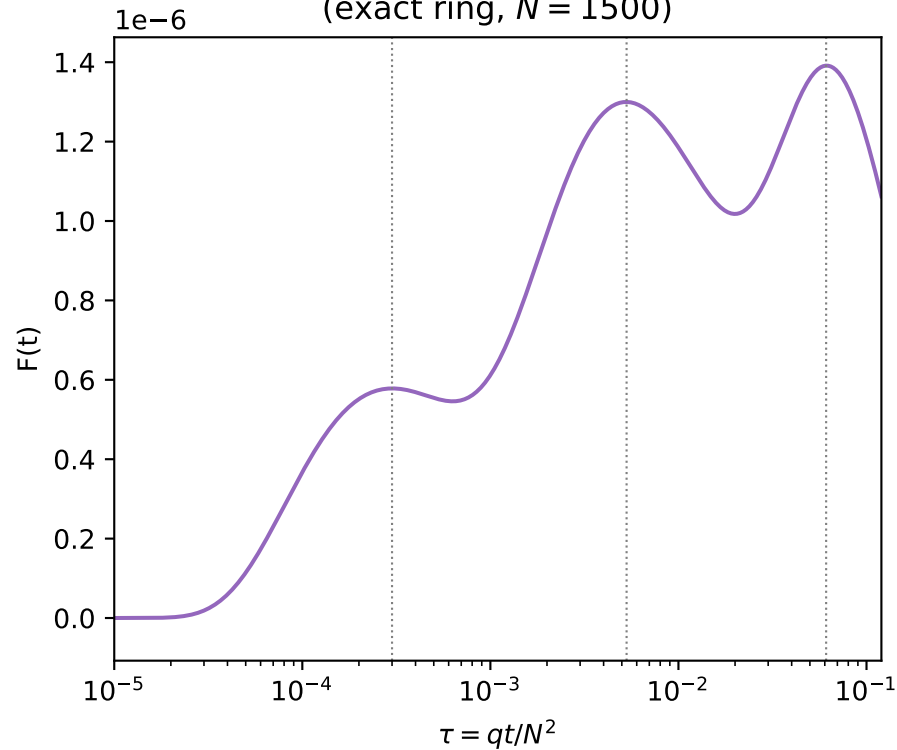
(b) master curve $\theta = \xi = 1/2$: second peak
annihilates at $b_c = 3.0764$



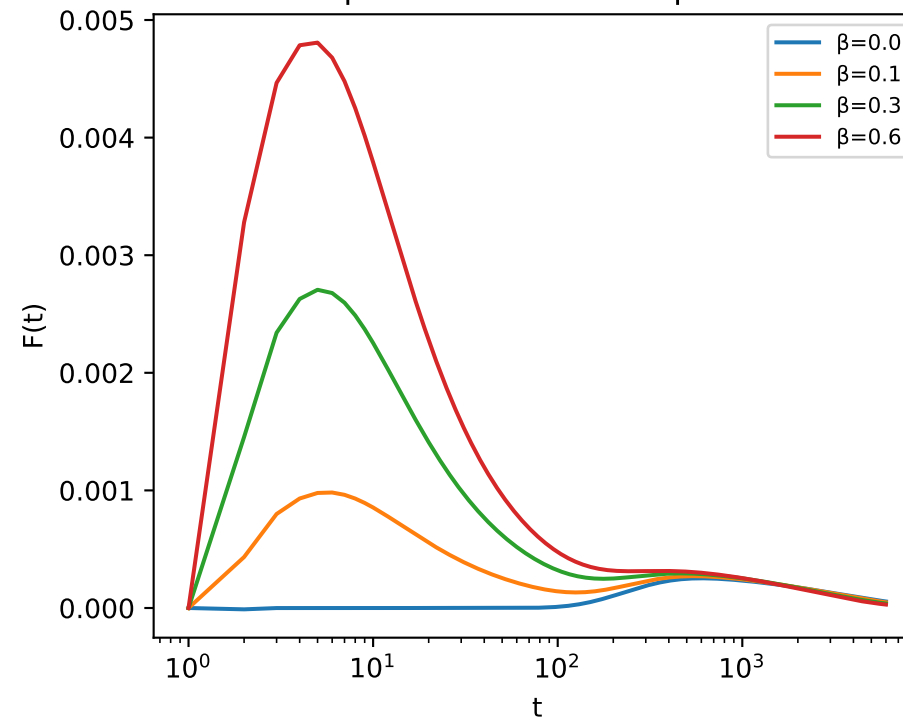
(c) normal-form fold scaling
($\delta^{1/2}$, $\delta^{3/2}$)



(d) two shortcuts, three peaks
(exact ring, $N = 1500$)



(e) 2D torus (31×31):
capture + late interior peaks



(f) 2D fold: late-peak prominence
 $\rightarrow 0$ at $\beta_c^{2D} \approx 0.69$

